

SC2000/CZ2100

Probability & Statistics

Week 2

Ch 2. Presenting Data

- Frequency tables and Charts
- Bar Charts
- Stem and Leaf Displays
- Histograms
- **Box Plots**
- **others**

- Box Plot:

Draw the Box Plot for the following data set (31 values).

14	15	16	16	17	17	17	17	17	18	
18	18	18	18	18	19	19	19	20	20	
20	20	20	20	21	21	22	23	24	24	29

Upper Hinge = 75th Percentile (3 rd quartile)	20
Lower Hinge = 25th Percentile (1 st quartile)	17
H-Spread = Upper Hinge - Lower Hinge (IQR)	3
Step = 1.5 x H-Spread	4.5
Upper Inner Fence = Upper Hinge + 1 Step	24.5
Lower Inner Fence = Lower Hinge - 1 Step	12.5

Upper Outer Fence = Upper Hinge + 2 Steps	29
Lower Outer Fence = Lower Hinge - 2 Steps	8
Upper Adjacent = Largest value $\leq$ Upper Inner Fence	24
Lower Adjacent = Smallest value $\geq$ Lower Inner Fence	14
Outside Value	Outlier = Value > upper adjacent or < lower adjacent
Far Out Value	
	29
	none

○ Outlier = 29

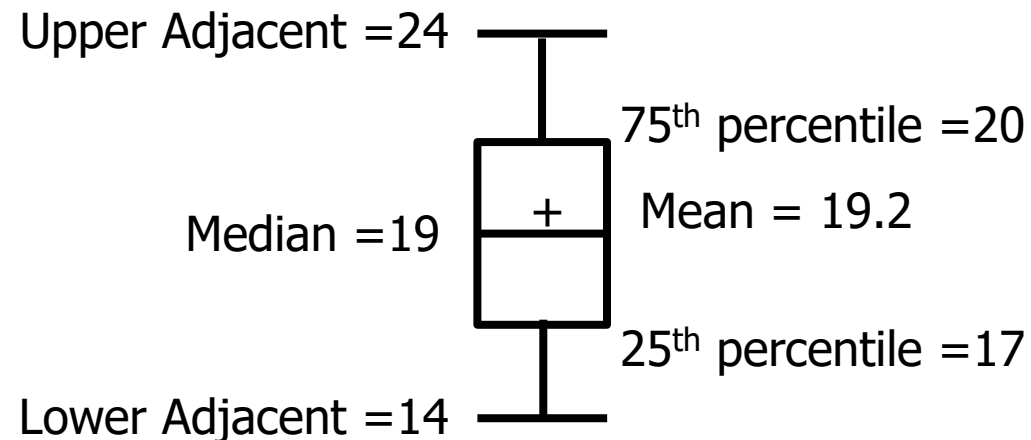
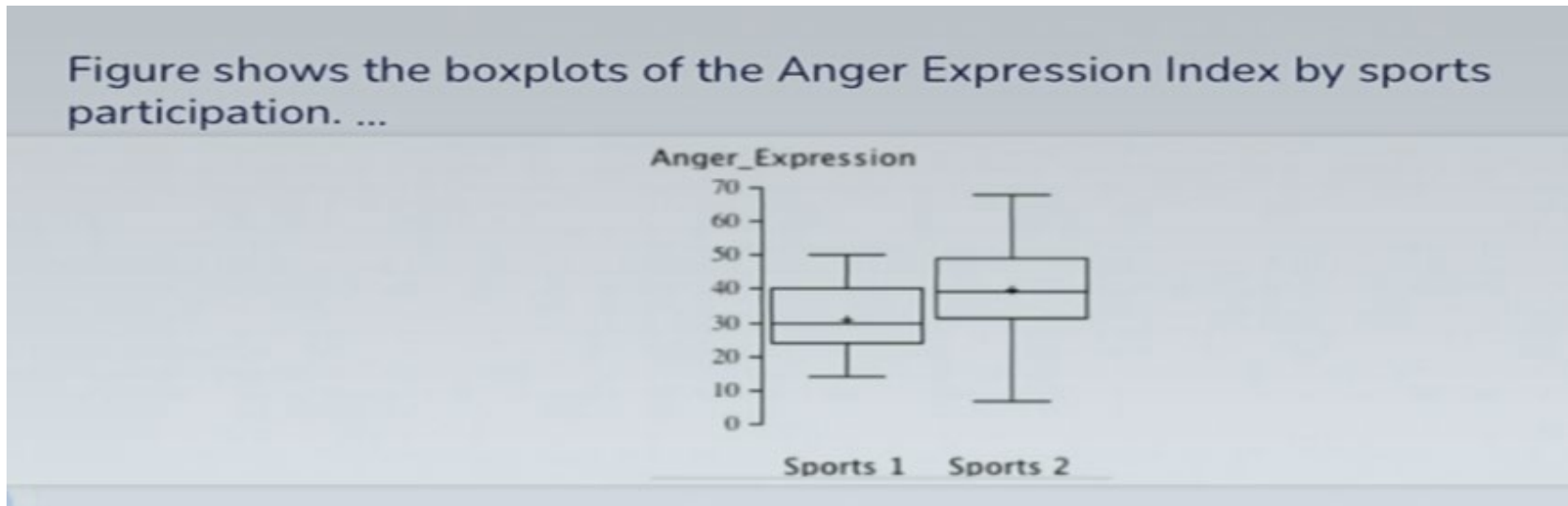
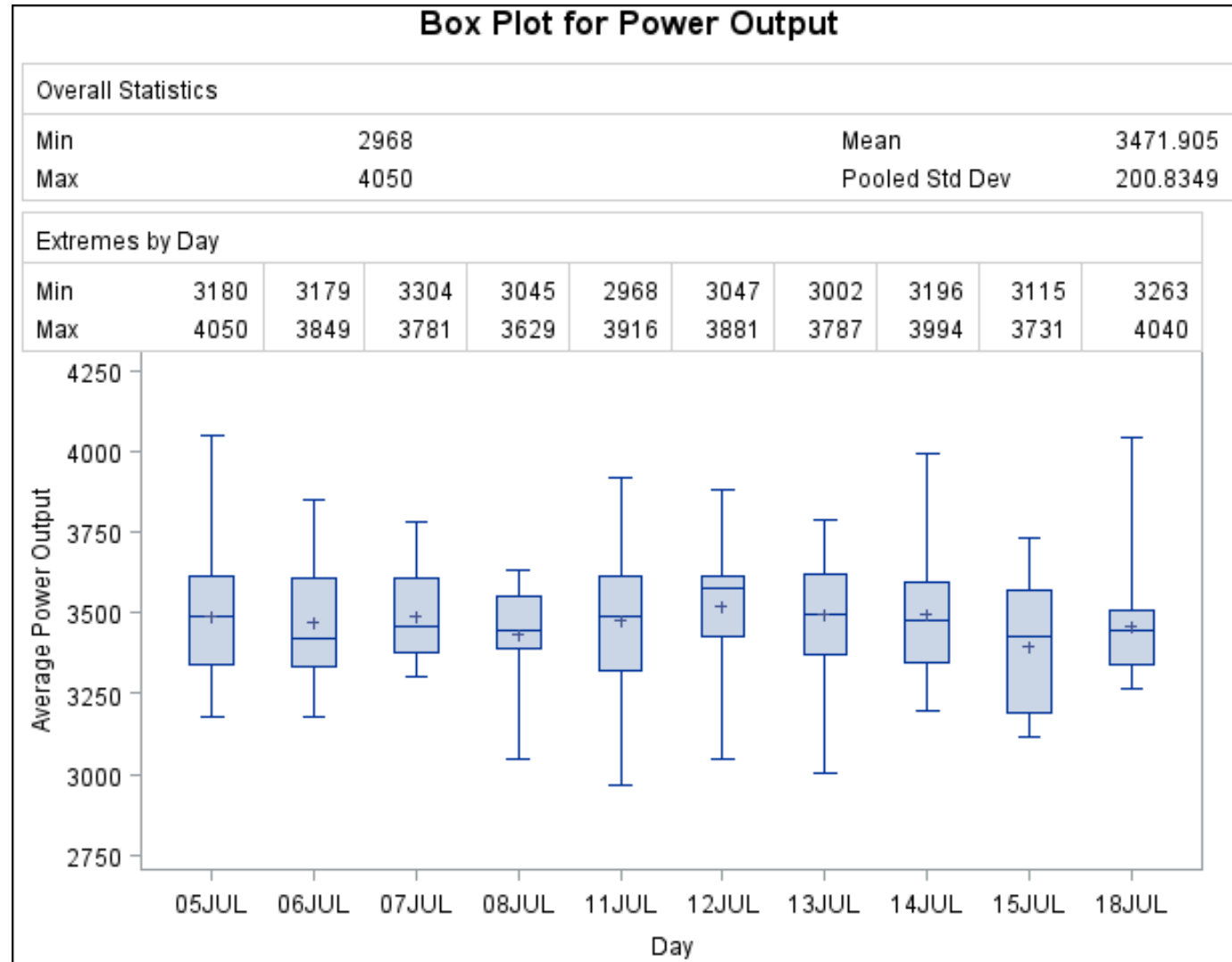


Figure shows the boxplots of the Anger Expression Index by sports participation. Does it look like there are any outliers? Which group reported expressing more anger?



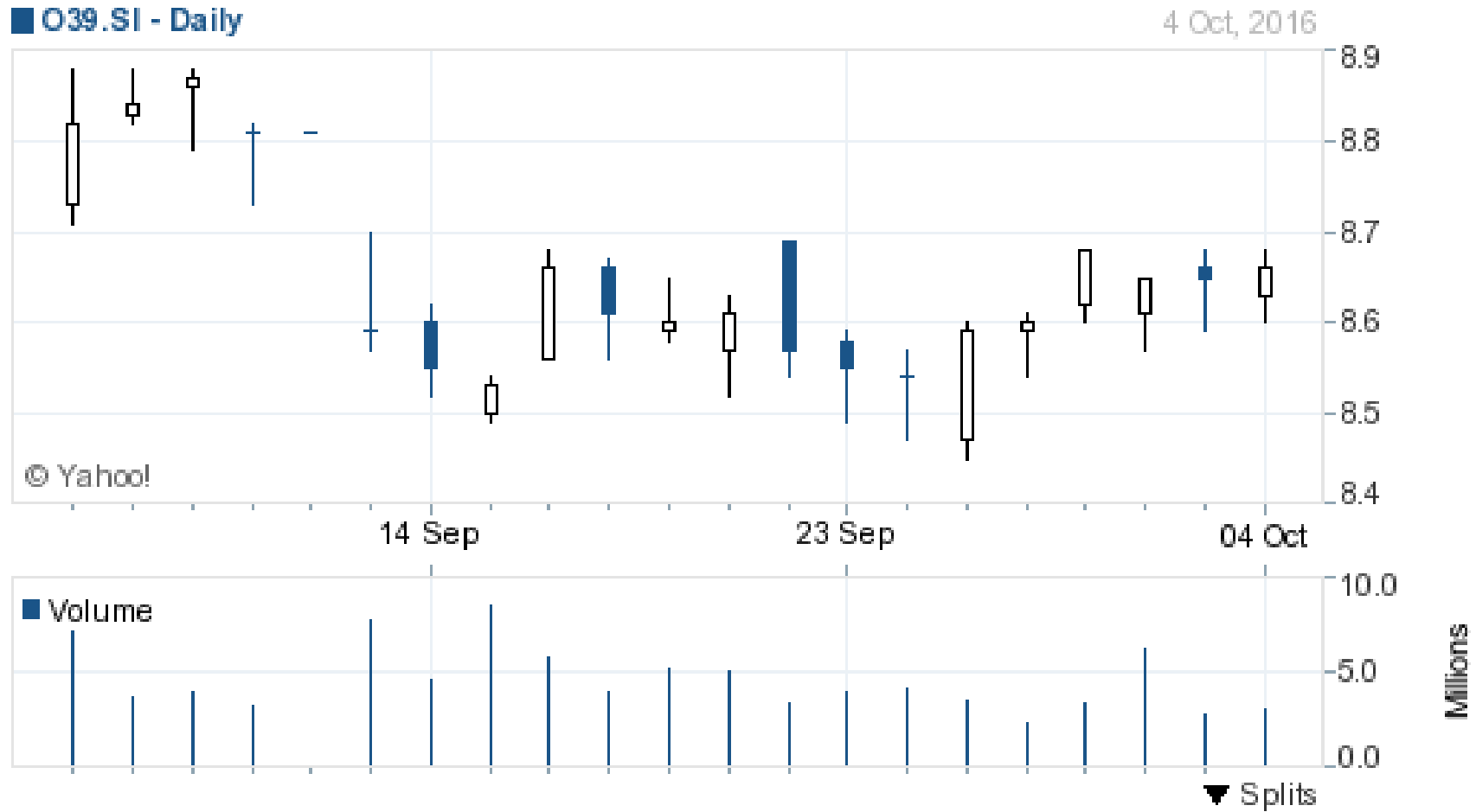
Practical example of Box Plot: Daily Power Output of an Electric Generator



Another practical example of Box Plot: Daily share prices

OCBC Bank

■ O39.SI - Daily



Ch 3. Summarizing Distributions

- Central Tendency: mean, median & mode
- Other Measures of Central Tendency
- Comparing Central Tendency
- Measures of Variability: Range, IQR, Variance
- Linear Transformation of variable
- Variance Sum Law I

■ Central Tendency

Summarizes a distribution by its central location

Different ways to define central tendency:

- Mean $\mu = \frac{\sum X}{N}$
- Median = 50th Percentile
- Mode = the value with the highest frequency
- Trimean = $\frac{25^{\text{th}} \text{ Percentile} + 2 * \text{Median} + 75^{\text{th}} \text{ Percentile}}{4}$
- Geometric mean = $(\prod X)^{\frac{1}{N}}$, where Π means to multiply
Eg: $\prod_{i=1}^5 X_i = X_1 * X_2 * X_3 * X_4 * X_5$
- Trimmed mean = mean for data with some higher and lower values removed

- Central Tendency:

Eg: Given the following data set, compute the mean, the median, the mode, the trimean, the geometric mean and the mean trimmed 18.2%.

1	3	4	4	4	5	5	7	8	9	31
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- Central Tendency – comparing various measures

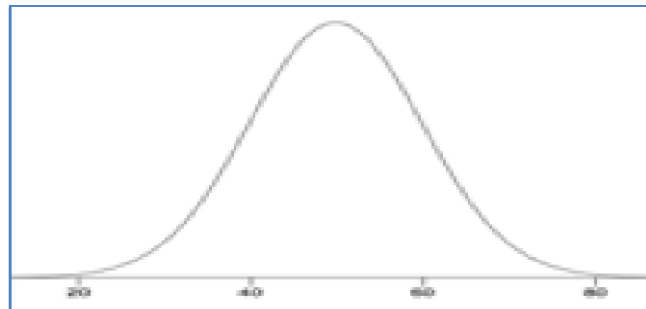
For symmetric distributions:

Mean = Median = Trimean = Trimmed mean
= Mode (except bimodal distr)

For skewed distributions:

Differences among the measures.

Example – the mean is typically higher than the median for a positive skewed distribution



Symmetric

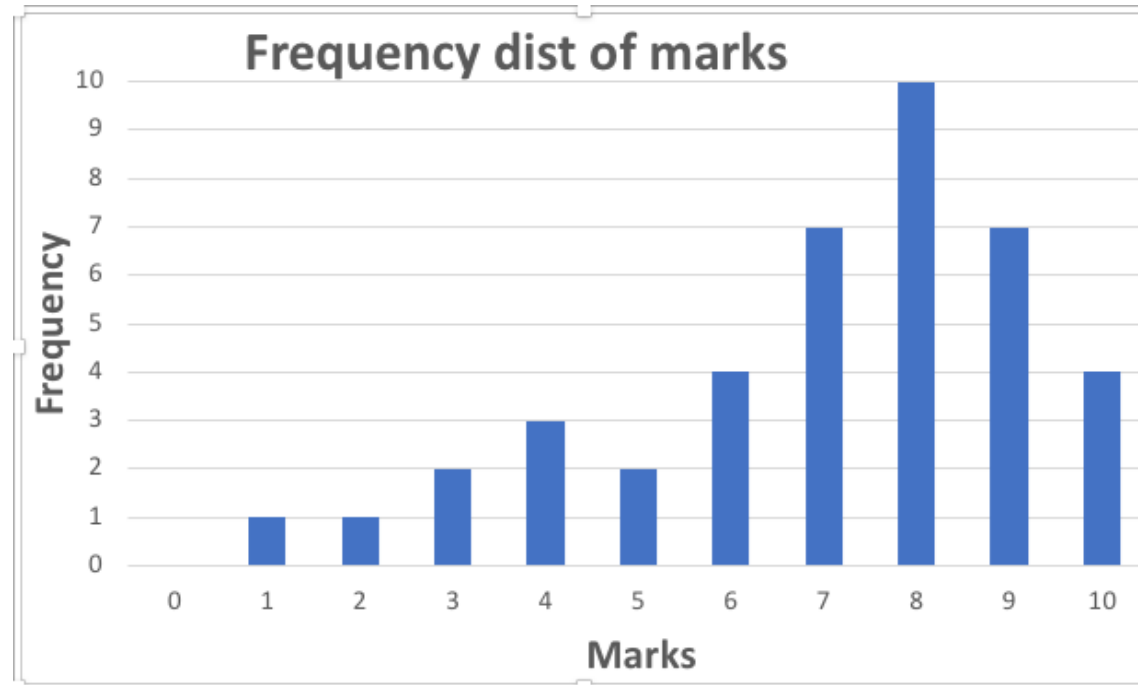
- Central Tendency – comparing various measures

Example:

Negative skewed

Calculate the following:

1. mode
2. median
3. trimean
4. mean



- Measures of Variability

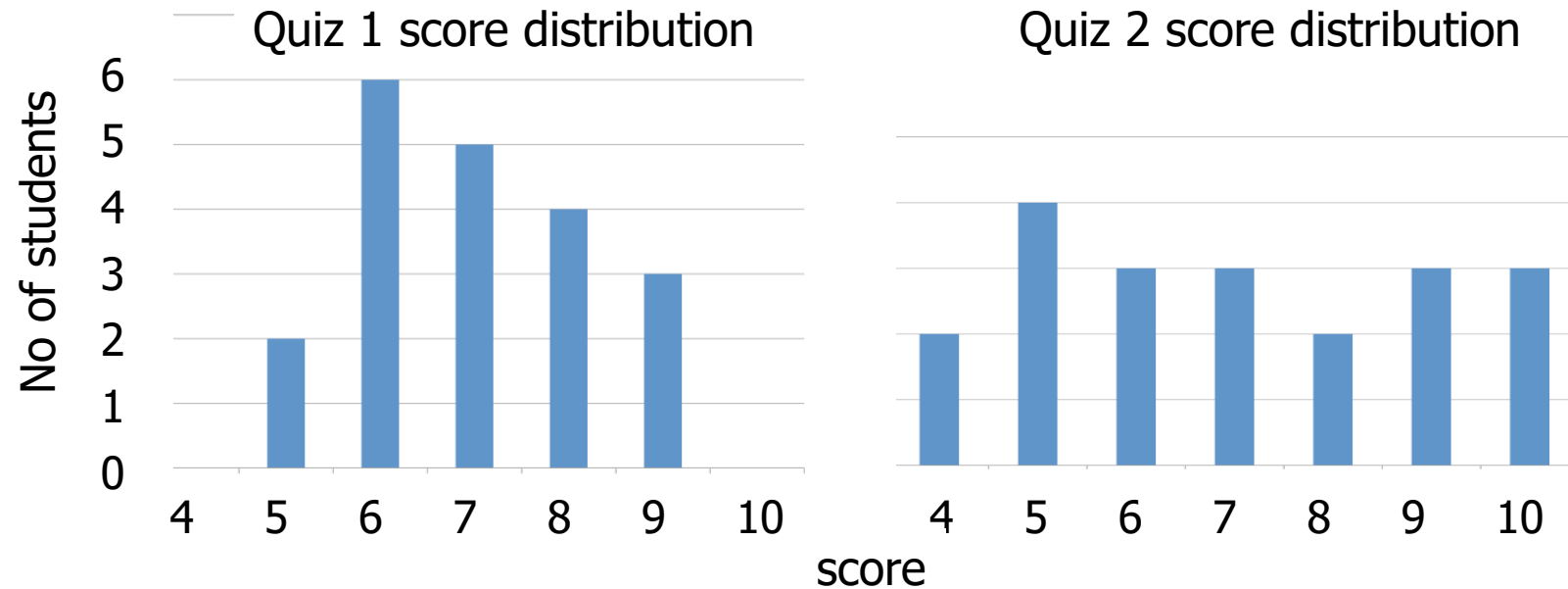
An indication of how spread out is the distribution

Frequently used measures of variability:

- Range = Highest value – Lowest value
- Interquartile Range IQR = 75th – 25th Percentile
- Variance $\sigma^2 = \frac{\sum(X - \mu)^2}{N}$
- Standard deviation = $\sqrt{\text{Variance}}$

- Variability:

Eg: Given the following 2 data set, compute the mean, the range, the IQR and the variance.



Mean μ		
Range		
IQR		
Var σ^2		

True or False questions:

1. A bimodal distribution has two modes and two medians. ①
2. The best way to describe a skewed distribution is to report the mean. ②
3. When plotted on the same graph, a distribution with a mean of 50 and a standard deviation of 10 will look more spread out than will a distribution with a mean of 60 and a standard deviation of 5. ③

Ch 3. Summarizing Distributions

- Central Tendency: mean, median & mode
- Other Measures of Central Tendency
- Comparing Central Tendency
- Measures of Variability: Range, IQR, Variance
- Linear Transformation of variable
- Variance Sum Law I

Compute mean and variance from the **population** of size N :

Population Mean $\mu = E[X] = \frac{\sum X}{N}$

$$\sigma^2 = E[X^2] - \mu^2$$

Population Variance $\sigma^2 = E[(X - \mu)^2] = \frac{\sum (X - \mu)^2}{N}$ or $\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$

Estimate mean and variance from a **sample** of size n :

Sample Mean $\bar{x} = \frac{\sum X}{n}$

Sample Variance $s^2 = \frac{\sum (X - \bar{x})^2}{n-1}$ or $\frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}$

Why $n-1$?

Suppose the denominator of s^2 is n , instead of $(n - 1)$:

$$s^2 = \frac{\sum (X - \bar{x})^2}{n} = \frac{1}{n} \left(\sum X^2 - \frac{(\sum X)^2}{n} \right)$$

For unbiased estimate, we expect the mean of s^2 to be equal to σ^2 :

$$\begin{aligned} E[s^2] &= E \left[\frac{1}{n} \left(\sum X^2 - \frac{(\sum X)^2}{n} \right) \right] \\ &= \frac{1}{n} \left(\sum \underbrace{E[X^2]}_{\sigma^2 + \mu^2 \text{ from the previous slide}} - \frac{E[(\sum X)^2]}{n} \right) \\ &= \frac{1}{n} \left(n \sigma^2 + n \mu^2 - \frac{E[(\sum X)^2]}{n} \right) \end{aligned}$$

$$\begin{aligned}
E[s^2] &= \frac{1}{n} \left(n \sigma^2 + n \mu^2 - \frac{E[(\sum X)^2]}{n} \right) \quad \leftarrow \begin{array}{l} \text{Let } Y = \sum X \\ E[Y^2] = \sigma_Y^2 + \mu_Y^2 \end{array} \\
&= \frac{1}{n} \left(n \sigma^2 + n \mu^2 - \frac{\text{Var}[\sum X] + (E[\sum X])^2}{n} \right) \\
&= \frac{1}{n} \left(n \sigma^2 + n \mu^2 - \frac{\sum \text{Var}[X] + (\sum E[X])^2}{n} \right) \\
&= \frac{1}{n} \left(n \sigma^2 + n \mu^2 - \frac{n \sigma^2 + (n \mu)^2}{n} \right) \\
&= \frac{1}{n} (n \sigma^2 + n \mu^2 - \sigma^2 - n \mu^2) = \frac{1}{n} (n - 1) \sigma^2
\end{aligned}$$

If the denominator is $(n - 1)$, then

the mean of s^2 is equal to σ^2 , i.e. $E[s^2] = \sigma^2$

- Linear transformation of variable
 - Transform data from one measurement scale to another

Eg. Consider a taxi trip from point A to B. The taxi service initial charge is \$3 and additional \$0.50 per km for the trip. Let y be the cost of the taxi ride and x be the distance travelled. We have:

$$y = 0.5x + 3$$

The mean of $y = 0.5$ (mean of x) + 3

The variance of $y = 0.5^2$ (variance of x)

- Variance Sum Law I

- Linear combination of 2 independent variables

Eg. Consider a couple A and B. A works x hrs/day and earns \$5/hr while B works y hrs/day and earns \$10/hr.

Total daily income t is: **Daily income difference d :**

$$t = 5x + 10y$$

$$d = 5x - 10y$$

Mean = 5 (mean of x) $-$ 10 (mean of y)

$$\text{Variance} = 5^2 \sigma_x^2 + 10^2 \sigma_y^2$$

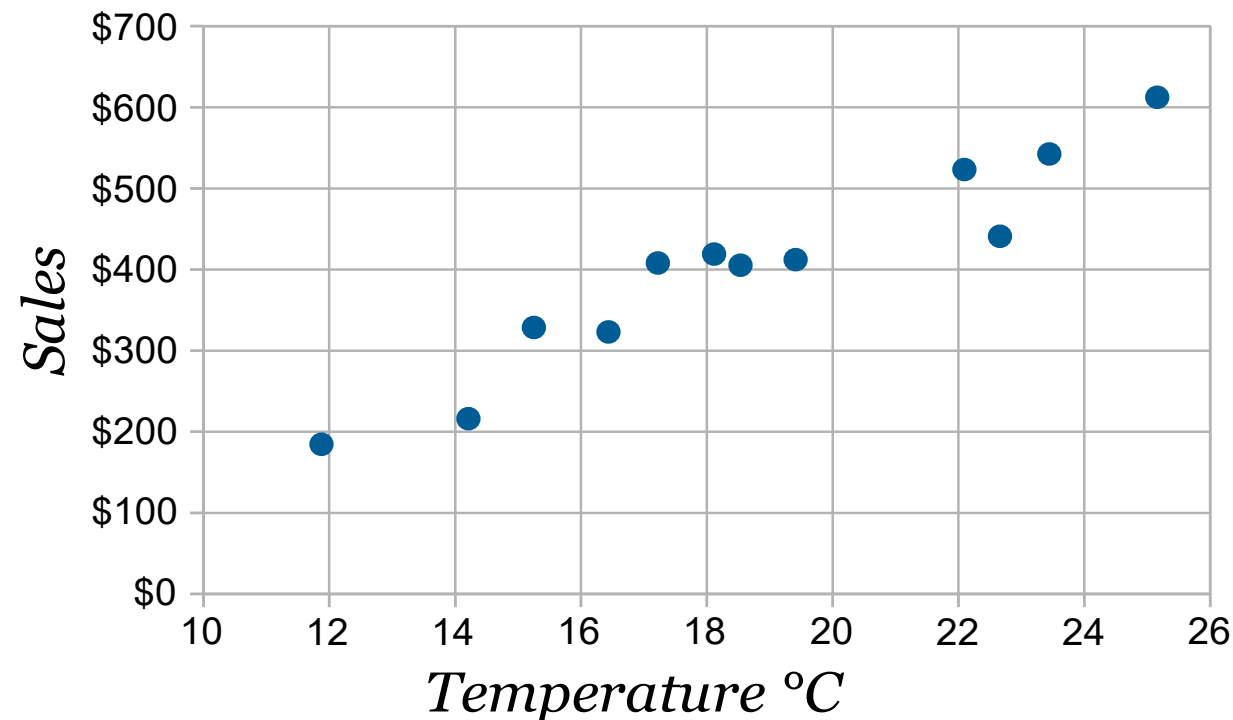
Ch 4. Bivariate Data

- Introduction to Bivariate Data
- Pearson Correlation and Covariance
- Properties of Person Correlation
- Variance Sum Law II

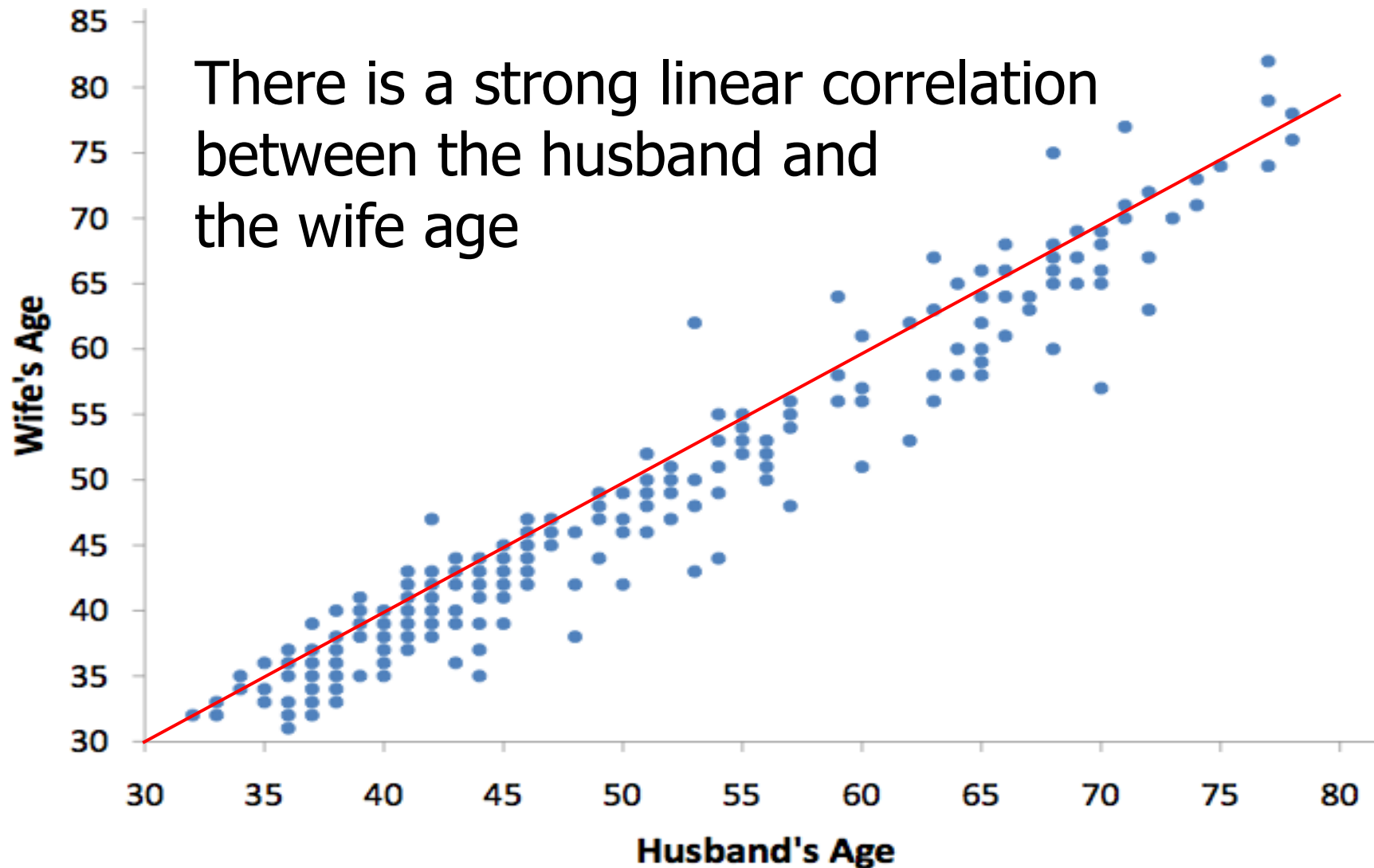
- Bivariate Data

- A dataset with a pair of variables which may be correlated to one another.

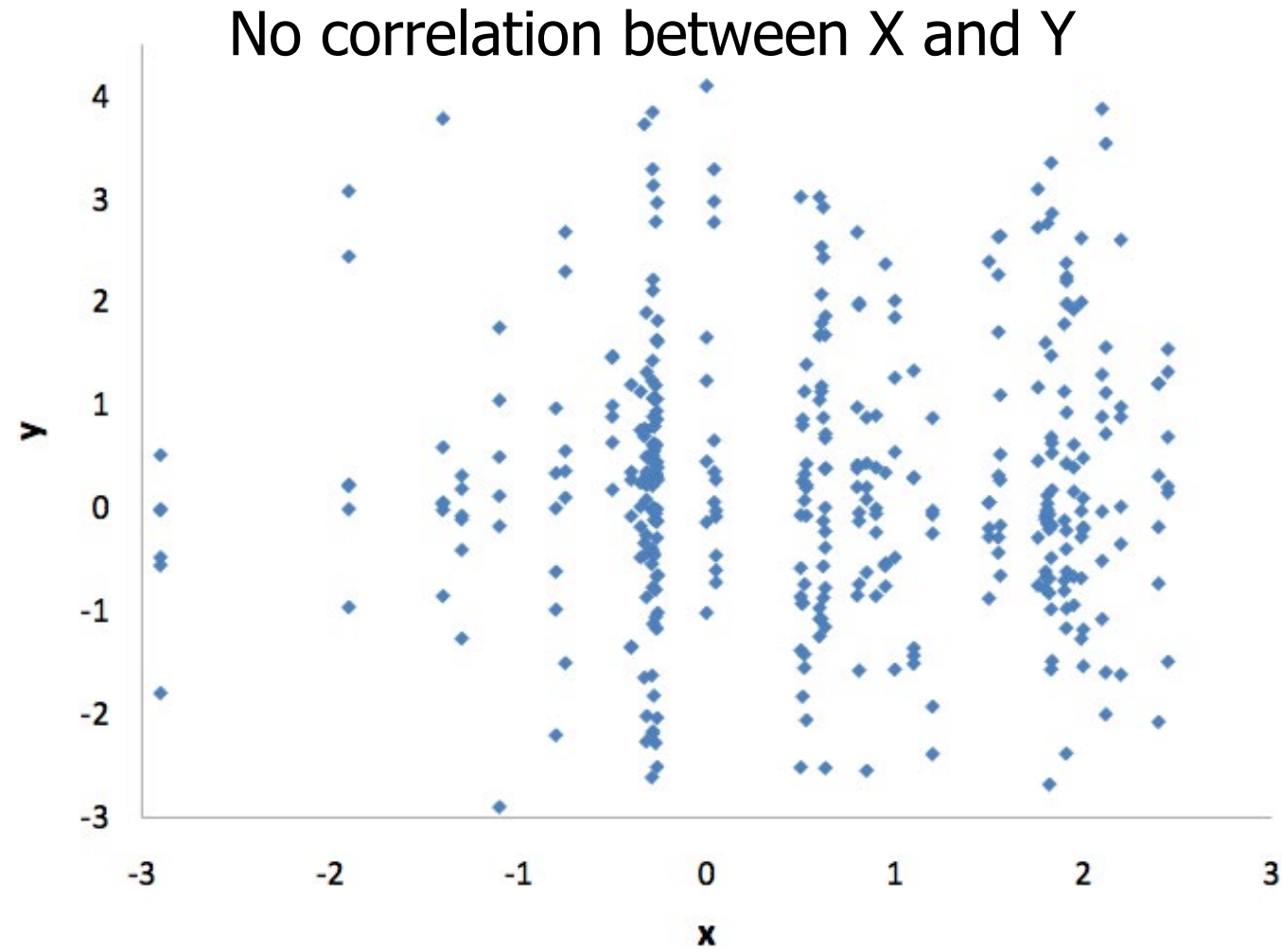
Eg: two variables – ice cream sales and temperature



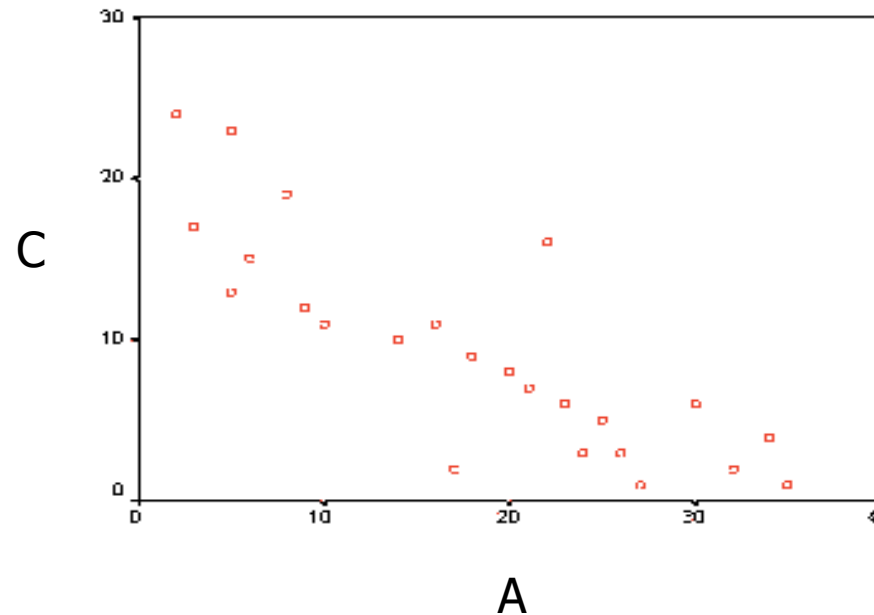
■ Bivariate Data



Eg: Correlation of variables X and Y



Question: Describe the relationship between variables A and C. Think of things these variables could represent in real life.



Negative relationship between A and C.
There is a negative relationship between price and quantity of the products that we buy.

■ Pearson Correlation ρ

- An indicator on the strength of the linear relationship between two variables.

Definition: $\rho = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$ Covariance of X and Y, denoted as cov(XY)

$$= \frac{E[XY] - \mu_X \mu_Y}{\sqrt{E[X^2] - (\mu_X)^2} \sqrt{E[Y^2] - (\mu_Y)^2}}$$

$$= \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}}$$

If $\mu_X = \mu_Y = 0$, then $\rho = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$

- Computation of Correlation based on a sample of size n

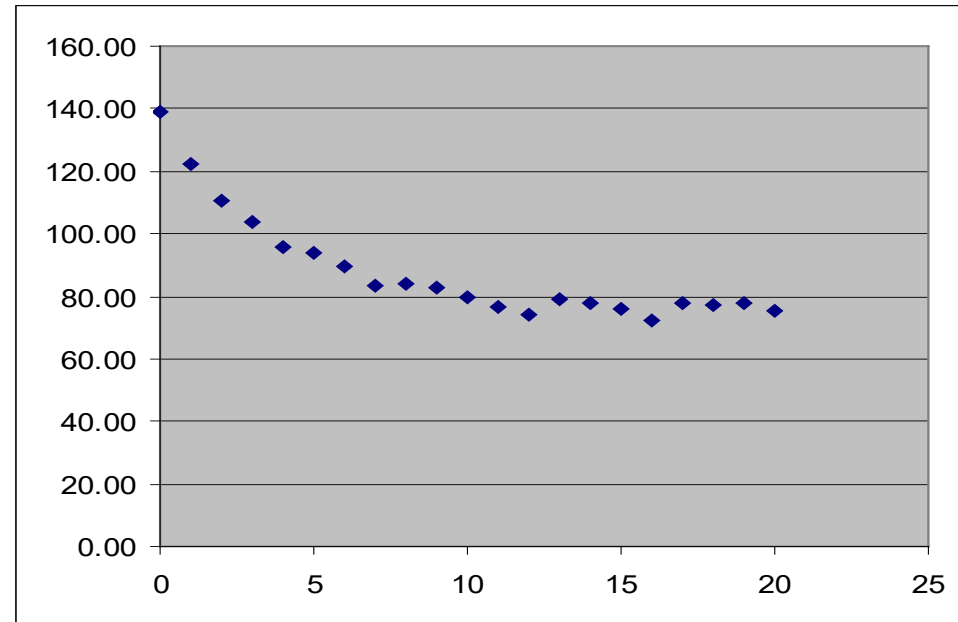
$$\text{cov}(XY) = \frac{1}{n-1} \sum (X - \bar{X})(Y - \bar{Y})$$

$$\text{Correlation } r = \frac{E[(X - \bar{X})(Y - \bar{Y})]}{s_X s_Y}$$

$$= \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{n}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{n}}}$$

$$\text{If } \bar{X} = \bar{Y} = 0, \text{ then } r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$$

Given the data shown in the figure, is it appropriate to use Pearson Correlation to describe the relationship between X and Y?



Eg: Given the data set, calculate σ_X , σ_Y , $\text{cov}(X,Y)$ and the Pearson Correlation.

X	2	5	6	8	9
Y	8	5	2	4	1

■ Properties of Correlation

- Value in the range of $[-1, +1]$
- Symmetric: correlation of X with Y
= correlation of Y with X
- Unaffected by linear transformations:
Correlation of Y with X
= correlation of Y with $A X + B$
where A and B are constants

Eg: If the correlation between weight (in pounds) and height (in feet) is 0.58, find:

- (a) the correlation between weight (in pounds) and height (in yards)
- (b) the correlation between weight (in kilograms) and height (in meters).

The correlation for both (a) and (b) is still 0.58 because linear transformations do not affect the value of Pearson's correlation, and both of the above instances are linear transformations.

Ch 4. Bivariate Data

- Introduction to Bivariate Data
- Pearson Correlation and Covariance
- Properties of Person Correlation
- **Variance Sum Law II**

- Variance Sum Law II

- Linear combination of 2 independent variables X and Y

Variance of $X \pm Y$: $\sigma_{x \pm y}^2 = \sigma_x^2 + \sigma_y^2$

- If the variables X and Y are correlated

Variance of $X \pm Y$: $\sigma_{x \pm y}^2 = \sigma_x^2 + \sigma_y^2 \pm 2\rho\sigma_x\sigma_y$

- For computation based on a sample

$$s_{x \pm y}^2 = s_x^2 + s_y^2 \pm 2rs_x s_y$$

Eg: Students took 2 parts of a test, each worth 50 points. Part A has a variance of 25, and Part B has a variance of 49. The correlation between the test scores is 0.6.

- (i) If the teacher adds the grades of the two parts together to form a final test grade, what would the variance of the final test grades be?
- (ii) What would the variance of Part A - Part B be?

$$\begin{aligned}\text{(i) } \text{Var} (A + B) &= 25 + 49 + 2*0.6*\sqrt{25}*\sqrt{49} \\ &= 116\end{aligned}$$

$$\begin{aligned}\text{(ii) } \text{Var} (A - B) &= 25 + 49 - 2*0.6*\sqrt{25}*\sqrt{49} \\ &= 32\end{aligned}$$